

Manuel Magana

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Software Engineer with Full-Stack and Data Engineering Experience

Software Engineer with 2.5 years at Genentech shipping Python tooling adopted by 20+ researchers. M.S. Computer Science from Northeastern University (2026), B.A. from UC Berkeley. Builds production backend systems in Python, Java, and Go with strong testing practices.

Education

Northeastern University

M.S. Computer Science

San Jose, CA

May 2026

Relevant Courses: Algorithms, Distributed Systems, Computer Networking, Software Engineering, Machine Learning, Deep Learning

University of California, Berkeley

B.A. Molecular and Cell Biology: Neurobiology

Berkeley, CA

May 2021

Relevant Courses: Data Structures, Machine Structure, Principles & Techniques of Data Science

Skills

Languages: Python, Go, Java, C, SQL, R, Bash

Frameworks & Tools: FastAPI, pytest, JUnit, Cucumber, Docker, Amazon Web Services (AWS), Git, Redis, MongoDB, PostgreSQL, DuckDB, Gradle, GitHub Actions, CI/CD

Libraries: pandas, PySpark, scikit-learn, PyTorch, TensorFlow/Keras

Work Experience

Genentech

Bioinformatics Scientist

San Francisco, CA

Nov 2022 – Jul 2025

- Co-designed and deployed a Python validation library with 2 engineers, containerized with Docker; reduced manual review from days to hours for 20+ researchers.
- Processed a 200GB+ mass spectrometry dataset using PySpark after it exceeded pandas' memory limits; converted raw data to Parquet for faster reads, enabling filtering and summarization at scale.
- Co-developed interactive dashboards used by 3+ research groups for experiment tracking and QC review.

Endless West

Junior Data Scientist

San Francisco, CA

Mar 2021 – Nov 2021

- Trained ML pipelines for sensory attribute prediction from chromatography data, with model outputs used to prioritize R&D synthesis targets.
- Developed Python web scrapers to extract and structure data from public databases, replacing hours of manual lookup with a searchable local dataset.

Projects

Trak | *Java, Swing, DuckDB, Redis, MongoDB, Gradle* | [GitHub](#)

May 2026 – Present

- Built a full-stack sprint planning tool in Java with a desktop GUI, CLI, and REST API (21 endpoints), featuring Google OAuth sign-in, email-based password recovery, and a first-run setup wizard.
- Implemented 5 interchangeable database backends (DuckDB, Redis, MongoDB, Parquet, JSON) that users can switch at runtime without code changes; shipped v1.0.0 as a native macOS installer.
- Wrote 254 automated tests (Cucumber BDD + JUnit) covering all 5 database backends, authentication flows, and API endpoints.

WhatToDo | *Python, FastAPI, Docker, AWS, PostgreSQL* | [GitHub](#)

Jan 2026 – Present

- Owned backend design for a 4-person team: a 3-model SQLAlchemy schema on AWS RDS with JWT auth and bcrypt hashing.
- Engineered a dual-provider AI engine (Claude + GPT-4o) with automatic failover between providers, keeping recommendations available even when one provider is down.
- Set up CI/CD with GitHub Actions and deployed containerized API to AWS (App Runner, ECR); achieved 96% test coverage.

Rate Limiter Service | *Go, Redis, Docker* | [GitHub](#)

Jan 2026

- Created a Go reverse proxy implementing fixed window and token bucket rate limiting with configurable per-user quotas.
- Leveraged Redis transactions for atomic token consumption to prevent race conditions; validated under 1,000+ concurrent requests.

Paxos Key-Value Store | *Java, RPC* | [GitHub](#)

Apr 2025

- Implemented a fault-tolerant distributed KV store using Paxos consensus with Proposer, Acceptor, and Learner roles over RPC.
- Validated under chaos testing: 10% acceptor failure rate, leader crashes, and network partitions while maintaining consensus.